In our example, we start out with a prior belief that this price is between 150k and 300k. The range from 150k to 300k, like your friend remarked, is quite big, so there's not much information contained in this initial prior belief—anything between 150k and 300k is possible. Now, the interesting thing happens when we *update* this range to a posterior belief in light of new information about either of two houses.

First, assuming that Alix's house in California comes at 250k, our posterior belief about our own house remains unchanged, from 150k to 300k. Again, this is because Alix's house is not relevant to ours, and the price of her house doesn't inform us about the quantity we're interested in.

Second, if the new information is that Alice's house, which is right next to our own, comes at 250k, then our posterior belief significantly changes from the prior to the 230k–270k range. Having Alice's house as a reference, we have updated our belief to be around the observed value, 250k, while narrowing down the range of our belief (going from a 150k difference to a 40k difference). This is a very reasonable thing to have, as the price of Alice's house is very informative with respect to the price of our house.

Figure 2.3 visualizes this entire process.

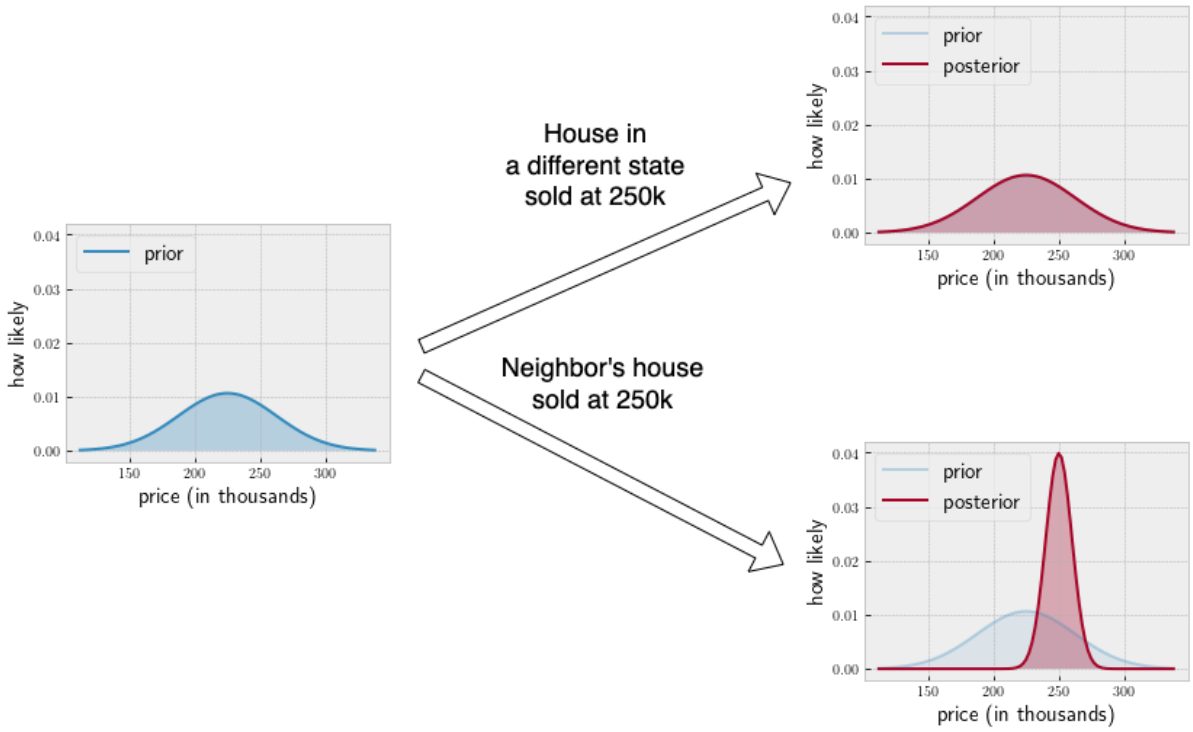


Figure 2.3 Updating the belief about the price of our house in a Bayesian way. Depending on how similar the house whose price was observed to our house, the posterior belief either stays the same or is drastically updated.

Note that the numbers above are not exact and are only there so that the example makes intuitive sense. However, we will see that by using a multivariate Gaussian distribution to model our belief, we can realize this intuitive update procedure in a quantifiable way. Further, with such a

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